



YOUR DIGITAL LAB

DECODE LABS REMOTE INTERNSHIP 2026

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- **INETRNSHIP DOMAIN:** DATA SCIENCE
 - **DATE:** 09/JUNE/2026
 - **TASK # 01(WEEK#01)**

Student Performance Analysis and Prediction

1. Introduction

This project focuses on analyzing student academic performance using data science techniques. The objective is to understand the factors affecting student exam scores and develop a machine learning model capable of predicting student performance.

The dataset contains 10,000 student records with information related to study habits, attendance, sleep patterns, internet usage, assignment completion, previous academic performance, and placement status.

2. Dataset Overview

Dataset Name: Student Performance Dataset

Total Records: 10,000

Total Features: 8

Features Included:

- Study Hours
- Attendance
- Sleep Hours
- Internet Usage
- Assignments Completed
- Previous Score
- Exam Score

- Placement Status
Target Variable:
 - Exam Score
Project Objective:
 - Analyze factors affecting student performance.
 - Identify important patterns and relationships.
 - Build a predictive model for exam score prediction.
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3. Data Collection and Understanding

The dataset was loaded and inspected to understand its structure and characteristics.

Dataset Shape:

- Rows: 10,000
- Columns: 8

Data Types:

- Numerical Features: 7
- Categorical Features: 1

Initial analysis confirmed that the dataset was well-structured and suitable for data analysis and machine learning tasks.

4. Data Cleaning and Preprocessing

Several preprocessing steps were performed to ensure data quality and consistency.

Cleaning Activities:

- Checked missing values
- Checked duplicate records
- Verified data types
- Validated column names
- Examined categorical values
- Verified numerical ranges

Results:

- No missing values found
- No duplicate records found
- All data types were correct
- Placement status contained valid categories
- No inconsistencies detected

The dataset was found to be clean and ready for analysis.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand relationships between variables and identify important patterns.

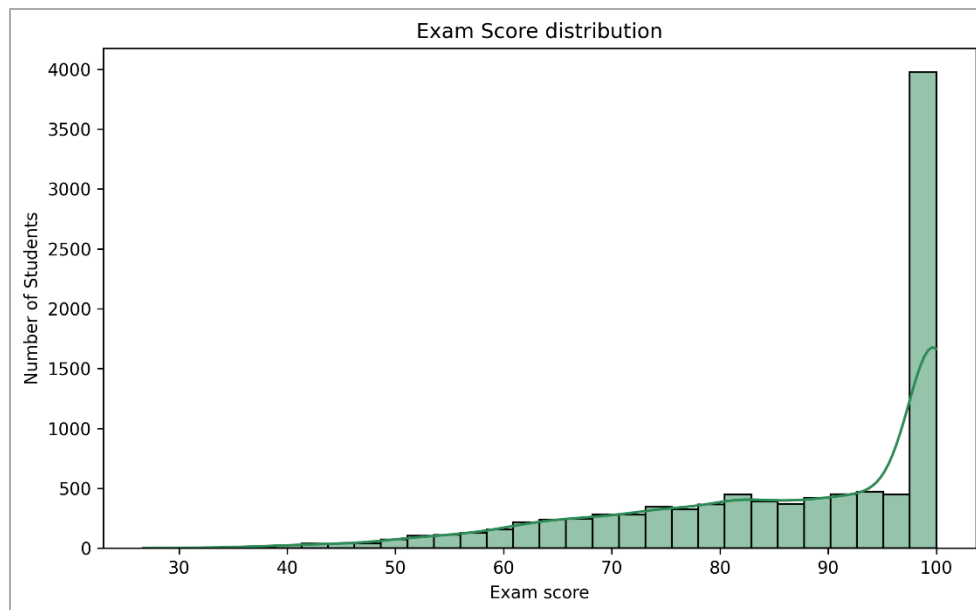
Key Findings:

1. Students study approximately 6 hours per day on average.
2. Average attendance is approximately 70%.
3. Average exam score is 86.70.
4. Students with higher study hours generally achieve better exam scores.
5. Previous academic performance shows a positive relationship with final exam scores.
6. Assignment completion contributes positively to academic performance.
7. Students with strong academic records are more likely to achieve higher scores.
8. Placement status appears associated with exam performance.
9. Academic behavior variables influence exam outcomes.
10. Multiple factors collectively contribute to student success.

6. Data Visualization

Several visualizations were created to better understand the dataset and communicate insights effectively.

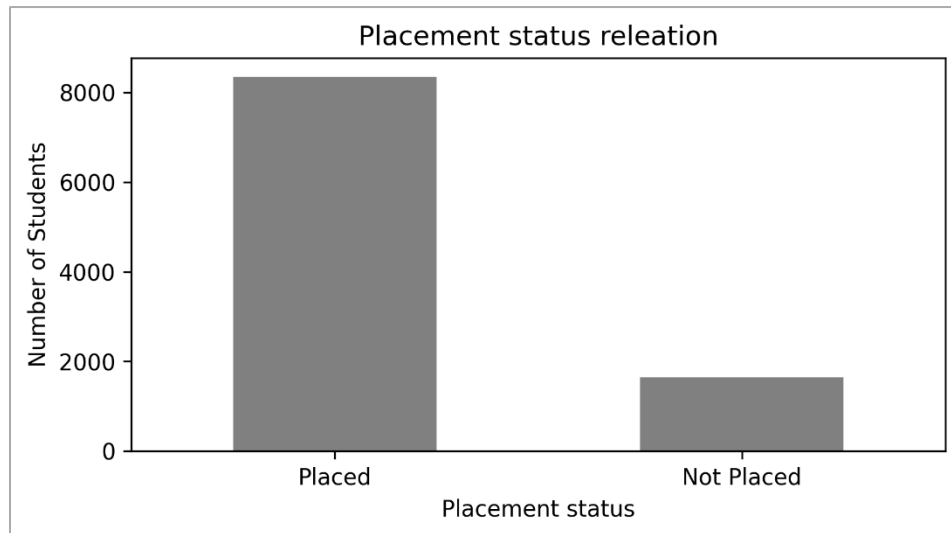
Chart 1: Exam Score Distribution



Purpose:

To understand how exam scores are distributed among students.

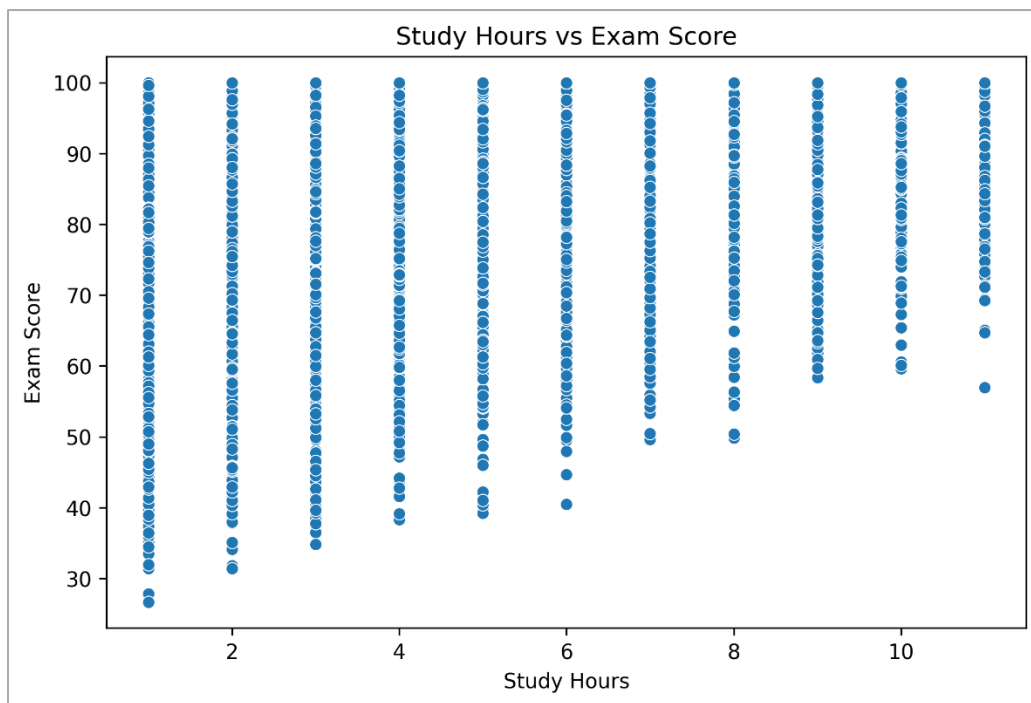
Chart 2: Placement Status Distribution



Purpose:

To compare the number of placed and non-placed students.

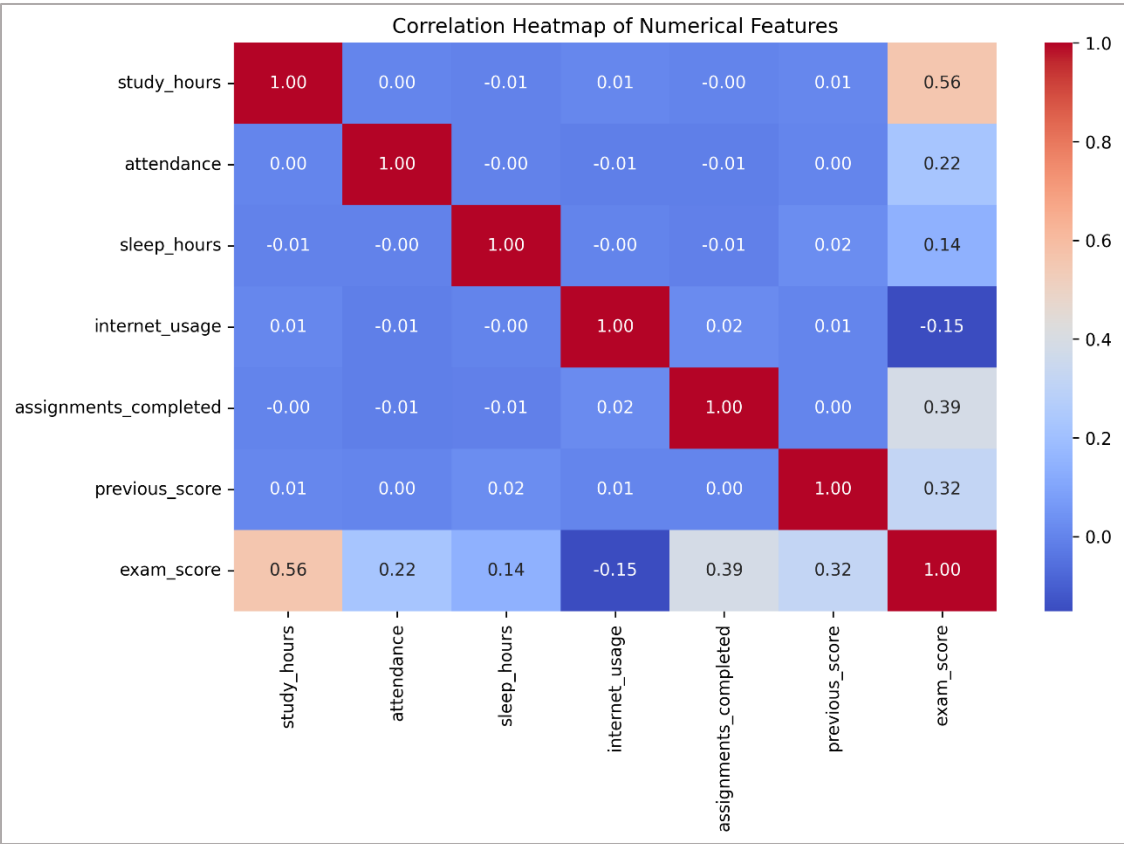
Chart 3: Study Hours vs Exam Score



Purpose:

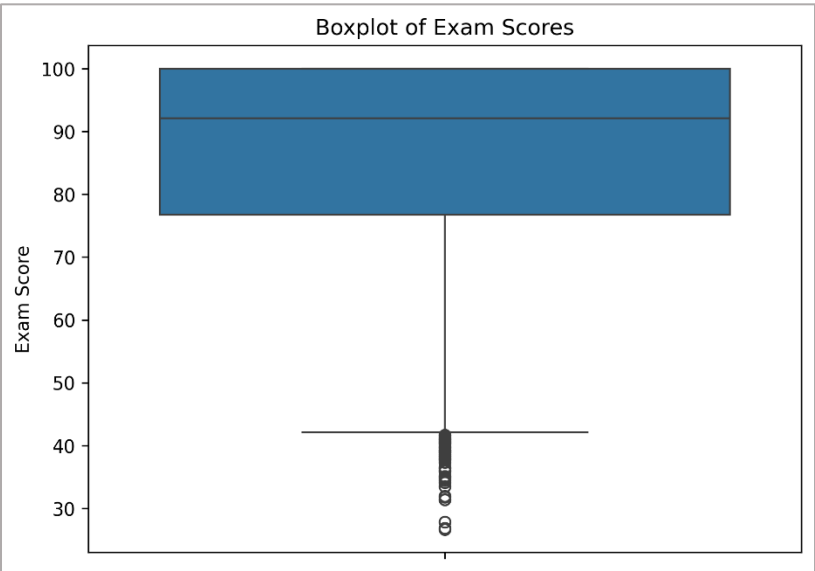
To analyze the relationship between study time and academic performance.

Chart 4: Correlation Heatmap



Purpose:
To identify relationships among numerical features.

Chart 5: Box Plot of Exam Scores



7. Machine Learning Model

A Linear Regression model was developed to predict student exam scores.

Features Used:

- Study Hours
- Attendance
- Sleep Hours
- Internet Usage
- Assignments Completed
- Previous Score

Target Variable:

- Exam Score

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

The model was trained using Linear Regression and evaluated on unseen data.

8. Model Performance

Evaluation Metrics:

Mean Absolute Error (MAE): 6.92

Root Mean Squared Error (RMSE): 8.79

R² Score: 0.663

Interpretation:

The model explains approximately 66.3% of the variation in exam scores. The results indicate that the model has reasonable predictive capability and can provide useful score predictions based on student characteristics.

9. Web Application Development

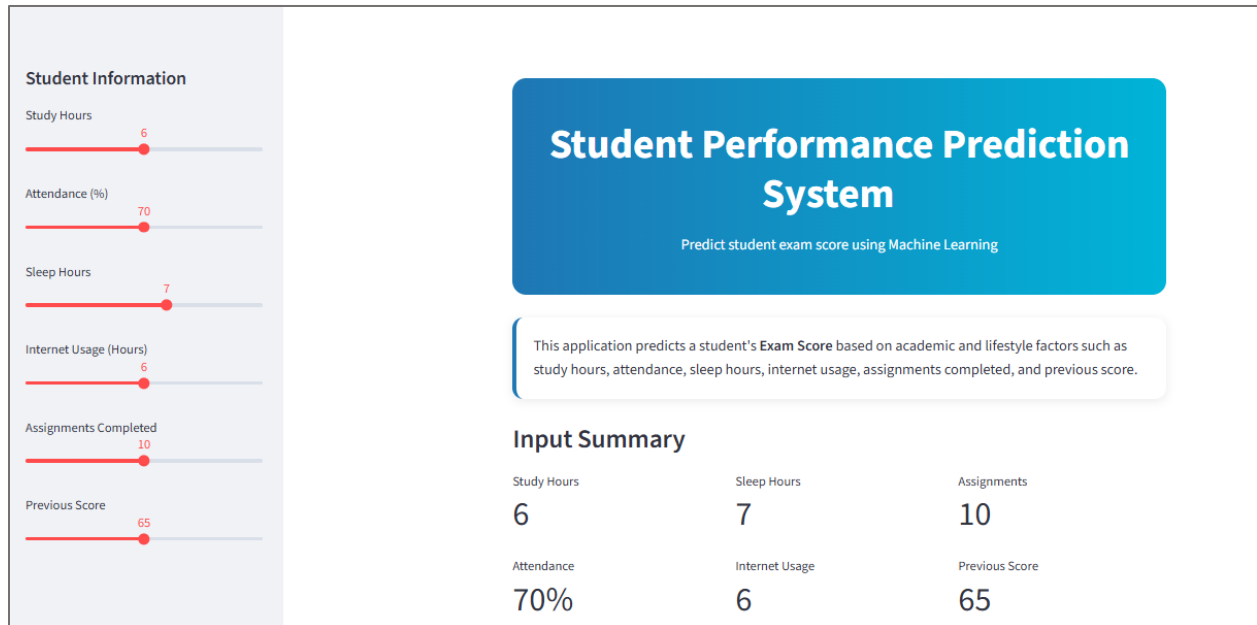
To enhance the project, a Streamlit-based web application was developed.

Application Features:

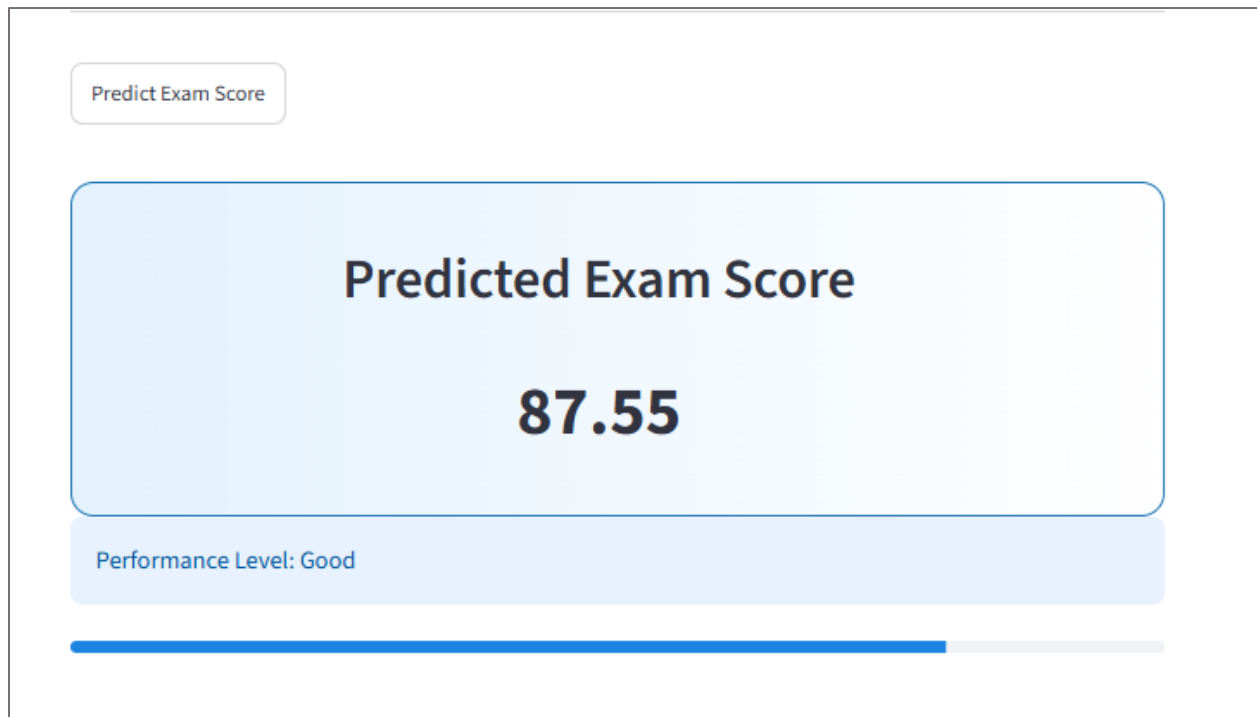
- User-friendly interface
- Interactive input controls
- Real-time score prediction
- Performance level classification
- Machine learning model integration

The application allows users to enter student information and instantly receive a predicted exam score.

Streamlit Application Screenshot



Prediction Result Screenshot



10. Conclusion

This project successfully applied data science techniques to analyze student performance and predict exam scores.

The project covered the complete data science workflow including:

- Data Collection
- Data Cleaning
- Exploratory Data Analysis
- Data Visualization
- Machine Learning
- Web Application Development

The developed Linear Regression model demonstrated satisfactory predictive performance and highlighted the influence of academic and behavioral factors on student success.

This project provides valuable insights into student performance and serves as a practical example of applying machine learning in the education domain.

Future Improvements

- Experiment with advanced machine learning algorithms
- Improve model accuracy through feature engineering
- Deploy the application online
- Add placement prediction functionality
- Develop a complete interactive dashboard

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Streamlit
- VS Code
- Jupyter Notebook

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